

Курсовая 08.04.2026

16. ~~линейное~~ ~~нема~~
N 1

$$Z = X_1 + 7X_2 \rightarrow \min$$

$$\begin{cases} 4X_1 + 7X_2 \geq 57 \\ 4X_1 + X_2 \geq 15 \\ 3X_1 + 11X_2 \geq 60 \\ X_i \geq 0, i=1,2 \end{cases}$$

$\Rightarrow$

$$-Z = -X_1 - 7X_2 \rightarrow \max$$

$$\begin{cases} 4X_1 + 7X_2 - X_3 = 57 \\ 4X_1 + X_2 - X_4 = 15 \\ 3X_1 + 11X_2 - X_5 = 60 \\ X_i \geq 0, i=1,5 \end{cases}$$

$$\left(\begin{array}{cccc|c} 4 & 7 & -1 & 0 & 0 & 57 \\ 4 & 1 & 0 & -1 & 0 & 15 \\ 3 & 11 & 0 & 0 & -1 & 60 \\ -1 & -7 & 0 & 0 & 0 & 1 \end{array} \right)$$

$\cdot -1$

$\Rightarrow$

$$\left(\begin{array}{cccc|c} -4 & -7 & 1 & 0 & 0 & -57 \\ -4 & -1 & 0 & 1 & 0 & -15 \\ -3 & -11 & 0 & 0 & 1 & -60 \\ -1 & -7 & 0 & 0 & 0 & 1 \end{array} \right)$$

$\Rightarrow$

Table 1

s.r.		x_1	x_2	x_3	x_4	x_5
x_3	57	-4	-7	1	0	0
x_4	15	-4	-1	0	1	0
x_5	60	-3	-11	0	0	1
-Z	0	4	7	0	0	0
CO		$-\frac{1}{3}$	$-\frac{7}{3}$			

$$\times (0, 0; -57, -15, 40)$$

Table 2

s.r.		x_1	x_2	x_3	x_4	x_5
x_3	23	0	5	1	0	3
x_4	65	0	2	0	1	2
x_1	20	1	0	0	0	3
-Z	20	0	$\frac{10}{3}$	0	0	$\frac{1}{3}$
CO						

$$\times (20, 0; 23, 65, 0)$$

$$-15 - \frac{-4 \cdot 20}{-3} = -15 + 80 = 65$$

$$-57 - \frac{-4 \cdot 20}{-3} = -57 + 80 = 23$$

$$0 - \frac{-10 \cdot 1}{-3} = -20$$

$$7 - \frac{1 \cdot -11}{-3} = 7 - \frac{11}{3} = \frac{21 - 11}{3} = \frac{10}{3}$$

$$0 - \frac{1 \cdot 1}{-3} = \frac{1}{3}$$

$$-Z_{\max} = Z(20, 0) \Rightarrow$$

$$\Rightarrow Z_{\min} = Z(20, 0) = 20 + 7 \cdot 0 = 20$$

$$\text{Optim: } Z_{\min} = Z(20, 0) = 20$$

1.2

$$Z = 6x_1 + x_2 + 2x_3 \rightarrow \min$$

$$\begin{cases} x_1 + x_2 + 5x_3 \geq 12 \\ x_1 + 8x_3 \geq 16 \\ 5x_1 + 2x_2 + x_3 \geq 10 \\ x_i \geq 0, i=1,2,3 \end{cases}$$

$$Z = -6x_1 - x_2 - 2x_3 \rightarrow \max$$

$$\begin{cases} 2x_1 + x_2 + 5x_3 - x_4 = 12 \\ x_1 + 8x_3 - x_5 = 16 \\ 5x_1 + 2x_2 + x_3 - x_6 = 10 \\ x_i \geq 0, i=1,6 \end{cases}$$

$$\left[\begin{array}{cccccc|c} 2 & 1 & 5 & -1 & 0 & 0 & 12 \\ 1 & 0 & 8 & 0 & -1 & 0 & 16 \\ 5 & 2 & 1 & 0 & 0 & -1 & 10 \end{array} \right] \quad \cdot -1$$

$$\Rightarrow \left[\begin{array}{cccccc|c} -2 & -1 & -5 & 1 & 0 & 0 & -12 \\ -1 & 0 & -8 & 0 & 1 & 0 & -16 \\ -5 & -2 & -1 & 0 & 0 & 1 & -10 \end{array} \right]$$

Table 1

i.k	1	x_1	x_2	x_3	x_4	x_5	x_6
x_4	-12	-2	-1	-5	1	0	0
x_5	-16	-1	0	-8	0	1	0
x_6	-10	-5	-2	-1	0	0	1
$-Z$	0	6	1	4	0	0	0
CO	12	-6		-2	12	16	10

X(0,0,0) -12, -16, -10

Table 2

i.k	1	x_1	x_2	x_3	x_4	x_5	x_6
x_4	-2	$-\frac{11}{8}$	-1	0	1	$\frac{5}{8}$	0
x_3	2	$\frac{1}{8}$	0	1	0	$-\frac{1}{8}$	0
x_6	-8	$-\frac{39}{8}$	-2	0	0	$-\frac{1}{8}$	1
Z	0	$\frac{49}{8}$	1	0	0	$\frac{1}{8}$	0
CO	12	$-\frac{19}{8}$	$-\frac{1}{8}$	12	16	-4	10

X(0,0,2) -2, -8, -10

$$-10 = \frac{-5 \cdot 4}{-8} = 10 + 10 = -2$$

$$-10 = \frac{-1 \cdot -4}{-8} = -10 + 2 = -8$$

$$0 = \frac{-4 \cdot 4}{-8} = -8$$

$$-2 = \frac{-5 \cdot -4}{-8} = -2 + \frac{5}{8} = \frac{-16 + 5}{8} = \frac{-11}{8}$$

$$-5 = \frac{-1 \cdot -1}{-8} = -5 + \frac{1}{8} = \frac{-40 + 1}{8} = \frac{-39}{8}$$

$$6 = \frac{-1 \cdot -1}{8} = 6 + \frac{1}{8} = \frac{49}{8}$$

$$5 = \frac{-5 \cdot 4}{-8} = 0 - \frac{5}{8} = \frac{-5}{8}$$

$$0 = \frac{1 \cdot -1}{-8} = \frac{1}{8}$$

Table 3:

		x_1	x_2	x_3	x_4	x_5	x_6	
20	x_1	2	1	0	0	1	1	1
20	x_2	2	1	0	1	0	1	1
20	x_3	4	1	1	0	0	1	1
-2	x_4	1	1	0	0	0	1	1
10	1	1	1	1	1	1	1	1

X(0, 4, 2, 2, 0, 0)

$$-2 = \frac{-1 \cdot -8}{-2} = -2 + 4 = 2$$

$$-8 = \frac{-8 \cdot 4}{-2} = -8 - 4 = -12$$

$$\frac{69}{8} = \frac{1 \cdot -39}{-2} = \frac{49}{8} + \frac{39}{16} = \frac{98 + 39}{16} = \frac{137}{16}$$

$$\frac{1}{2} = \frac{1 \cdot 1}{-2} = \frac{1}{-2} + \frac{1}{2} = 1$$

$$0 = \frac{1 \cdot 1}{-2} = \frac{1}{-2}$$

$$Z_{\max} = Z(0; 4; 2; 2; 0; 0)$$

$$Z_{\min} = Z(0; 4; 2; 2; 0; 0) =$$

$$= 6 \cdot 0 + 4 + 9 \cdot 2 = 4 + 18 = 22$$

$$\text{Geben: } Z_{\min} = Z(0; 4; 2) = 12$$

3

$$Z = 5x_1 + 2x_2 + 3x_3 \rightarrow \min$$

$$2x_1 + 3x_2 + 4x_3 \geq 10$$

$$Z = 5x_1 + 2x_2 + 3x_3 \rightarrow \min$$

$$\begin{cases} x_1 + x_2 + x_3 \geq 10 \\ 2x_1 + 3x_2 + 4x_3 \geq 20 \\ 5x_2 + 6x_3 \geq 30 \\ x_i \geq 0; i=1,2,3 \end{cases}$$

$$2x_1 + 3x_2 + 4x_3 \geq 20$$

$$5x_2 + 6x_3 \geq 30$$

$$x_i \geq 0; i=1,2,3$$

$$-Z = -5x_1 - 2x_2 - 3x_3$$

$$\begin{cases} x_1 + x_2 + x_3 - x_4 = 10 \\ 2x_1 + 3x_2 + 4x_3 - x_5 = 20 \\ 5x_2 + 6x_3 - x_6 = 30 \\ x_i \geq 0; i=1,6 \end{cases}$$

$$2x_1 + 3x_2 + 4x_3 - x_5 = 20$$

$$5x_2 + 6x_3 - x_6 = 30$$

$$x_i \geq 0; i=1,6$$

$$x_1 \rightarrow \begin{pmatrix} -1 & -2 & -1 & 1 & 0 & 0 & -10 \\ -2 & -3 & -4 & 0 & 1 & 0 & -20 \\ 30 & -5 & -6 & 0 & 0 & 1 & -30 \end{pmatrix}$$

Table 1:

		x_1	x_2	x_3	x_4	x_5	x_6
x_1	-10	-1	-2	-1	1	0	0
x_5	-20	-2	-3	-4	0	1	0
x_6	-30	0	-5	-6	0	0	1
Z	0	5	2	3	0	0	0
CG	///	-2	-3	-4	///	///	///

$$x_1 \rightarrow (0; 0; 0; 10; 20; 30)$$

Task 2:

	x_1	x_2	x_3	x_4	x_5	x_6
x_1	1	0	0	0	0	0
x_2	0	1	0	0	0	0
x_3	0	0	1	0	0	0
x_4	0	0	0	1	0	0
x_5	0	0	0	0	1	0
x_6	0	0	0	0	0	1
x_7	0	0	0	0	0	0
x_8	0	0	0	0	0	0
x_9	0	0	0	0	0	0
x_{10}	0	0	0	0	0	0
x_{11}	0	0	0	0	0	0
x_{12}	0	0	0	0	0	0
x_{13}	0	0	0	0	0	0
x_{14}	0	0	0	0	0	0
x_{15}	0	0	0	0	0	0
x_{16}	0	0	0	0	0	0
x_{17}	0	0	0	0	0	0
x_{18}	0	0	0	0	0	0
x_{19}	0	0	0	0	0	0
x_{20}	0	0	0	0	0	0
x_{21}	0	0	0	0	0	0
x_{22}	0	0	0	0	0	0
x_{23}	0	0	0	0	0	0
x_{24}	0	0	0	0	0	0
x_{25}	0	0	0	0	0	0
x_{26}	0	0	0	0	0	0
x_{27}	0	0	0	0	0	0
x_{28}	0	0	0	0	0	0
x_{29}	0	0	0	0	0	0
x_{30}	0	0	0	0	0	0
x_{31}	0	0	0	0	0	0
x_{32}	0	0	0	0	0	0
x_{33}	0	0	0	0	0	0
x_{34}	0	0	0	0	0	0
x_{35}	0	0	0	0	0	0
x_{36}	0	0	0	0	0	0
x_{37}	0	0	0	0	0	0
x_{38}	0	0	0	0	0	0
x_{39}	0	0	0	0	0	0
x_{40}	0	0	0	0	0	0
x_{41}	0	0	0	0	0	0
x_{42}	0	0	0	0	0	0
x_{43}	0	0	0	0	0	0
x_{44}	0	0	0	0	0	0
x_{45}	0	0	0	0	0	0
x_{46}	0	0	0	0	0	0
x_{47}	0	0	0	0	0	0
x_{48}	0	0	0	0	0	0
x_{49}	0	0	0	0	0	0
x_{50}	0	0	0	0	0	0
x_{51}	0	0	0	0	0	0
x_{52}	0	0	0	0	0	0
x_{53}	0	0	0	0	0	0
x_{54}	0	0	0	0	0	0
x_{55}	0	0	0	0	0	0
x_{56}	0	0	0	0	0	0
x_{57}	0	0	0	0	0	0
x_{58}	0	0	0	0	0	0
x_{59}	0	0	0	0	0	0
x_{60}	0	0	0	0	0	0
x_{61}	0	0	0	0	0	0
x_{62}	0	0	0	0	0	0
x_{63}	0	0	0	0	0	0
x_{64}	0	0	0	0	0	0
x_{65}	0	0	0	0	0	0
x_{66}	0	0	0	0	0	0
x_{67}	0	0	0	0	0	0
x_{68}	0	0	0	0	0	0
x_{69}	0	0	0	0	0	0
x_{70}	0	0	0	0	0	0
x_{71}	0	0	0	0	0	0
x_{72}	0	0	0	0	0	0
x_{73}	0	0	0	0	0	0
x_{74}	0	0	0	0	0	0
x_{75}	0	0	0	0	0	0
x_{76}	0	0	0	0	0	0
x_{77}	0	0	0	0	0	0
x_{78}	0	0	0	0	0	0
x_{79}	0	0	0	0	0	0
x_{80}	0	0	0	0	0	0
x_{81}	0	0	0	0	0	0
x_{82}	0	0	0	0	0	0
x_{83}	0	0	0	0	0	0
x_{84}	0	0	0	0	0	0
x_{85}	0	0	0	0	0	0
x_{86}	0	0	0	0	0	0
x_{87}	0	0	0	0	0	0
x_{88}	0	0	0	0	0	0
x_{89}	0	0	0	0	0	0
x_{90}	0	0	0	0	0	0
x_{91}	0	0	0	0	0	0
x_{92}	0	0	0	0	0	0
x_{93}	0	0	0	0	0	0
x_{94}	0	0	0	0	0	0
x_{95}	0	0	0	0	0	0
x_{96}	0	0	0	0	0	0
x_{97}	0	0	0	0	0	0
x_{98}	0	0	0	0	0	0
x_{99}	0	0	0	0	0	0

$$-10 - \frac{-1 \cdot -30}{-5} = -10 + 6 = -4$$

$$-20 - \frac{-3 \cdot -30}{-5} = -20 + 18 = -2$$

$$0 - \frac{6 \cdot -2}{-5} = -12$$

$$-1 - \frac{-1 \cdot -6}{-5} = -1 + \frac{6}{5} = \frac{1}{5}$$

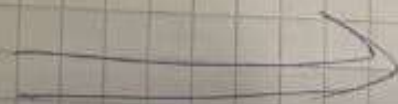
$$-4 - \frac{-3 \cdot -6}{-5} = -4 + \frac{18}{5} = \frac{-20 + 18}{5} = \frac{-2}{5}$$

$$9 - \frac{2 \cdot -6}{-5} = 9 - \frac{12}{5} = \frac{45 - 12}{5} = \frac{33}{5}$$

$$0 - \frac{-1 \cdot 1}{-5} = -\frac{1}{5}$$

$$0 - \frac{-3 \cdot 1}{-5} = -\frac{3}{5}$$

$$0 - \frac{1 \cdot 2}{-5} = \frac{2}{5}$$



Problem 3

		x_1	x_2	x_3	x_4	x_5	x_6
x_1	20	1	0	1	2	0	1
x_2	10	1	0	1	1	1	0
x_3	10	1	1	1	1	0	0
x_4	20	3	0	7	2	0	0
x_5	10	1	1	1	1	1	1

$$-2 - \frac{-4}{5} \cdot \frac{3}{5} = -2 + \frac{12}{25} = -10$$

$$6 - \frac{-4}{5} \cdot \frac{1}{5} = 6 + \frac{4}{25} = 10$$

$$5 - \frac{-1}{5} \cdot \frac{2}{5} = 5 - 2 = 3$$

$$\frac{33}{5} - \frac{1}{5} \cdot \frac{2}{5} = \frac{33}{5} + \frac{2}{5} = \frac{35}{5} = 7$$

$$0 - 1 \cdot \frac{2}{5} = 0 + 2 = 2$$

$$-Z_{\max} = -Z(0; 10; 0) = -20$$

$$Z_{\min} = Z(0; 10; 0) = -5 \cdot 0$$

$$= 5 \cdot 0 + 2 \cdot 10 + 9 \cdot 0 = 20$$

Answer: $Z_{\min} = Z(0; 10; 0) = 20$

Линейное программирование 1.6

$$Z = X_1 + 9X_2 + 10X_3 \rightarrow \min$$

$$\begin{cases} 2X_1 + 3X_2 + 5X_3 \geq 24 \\ 3X_1 + X_2 + X_3 \geq 12 \\ X_1 + 9X_2 + 5X_3 \geq 20 \\ X_i \geq 0, i = 1, 2, 3 \end{cases}$$

$$-Z = -X_1 - 9X_2 - 10X_3 \rightarrow \max$$

$$\begin{cases} 2X_1 + 3X_2 + 5X_3 - X_4 = 24 \\ 3X_1 + X_2 + X_3 - X_5 = 12 \\ X_1 + 9X_2 + 5X_3 - X_6 = 20 \\ X_i \geq 0, i = 1, 6 \end{cases}$$

$$\left(\begin{array}{cccccc|c} -2 & -3 & -5 & 1 & 0 & 0 & -24 \\ -3 & -1 & -1 & 0 & 1 & 0 & -12 \\ -1 & -9 & -5 & 0 & 0 & 1 & -20 \end{array} \right)$$

Шаг 1:

ЭЛ		X_1	X_2	X_3	X_4	X_5	X_6
X_4	-24	-2	-3	-5	1	0	0
X_5	-12	-3	-1	-1	0	1	0
X_6	-20	-1	-9	-5	0	0	1
$-Z$	0	1	9	10	0	0	0
CB	1	-1	-9	-5	0	0	0

Exercise 2:

		x_1	x_2	x_3	x_4	x_5	x_6
x_1	1	1	$\frac{5}{2}$	$\frac{5}{2}$	$-\frac{1}{2}$	0	0
x_2	10	1	$\frac{5}{2}$	$\frac{5}{2}$	$-\frac{1}{2}$	0	0
x_3	24	0	$\frac{3}{2}$	$\frac{13}{2}$	$-\frac{3}{2}$	1	0
x_4	8	0	$-\frac{5}{2}$	$-\frac{5}{2}$	$\frac{1}{2}$	0	1
x_5	-12	0	$\frac{15}{2}$	$\frac{15}{2}$	$\frac{1}{2}$	0	0
x_6	12	1	3	3	-1	0	0

$$-12 - \frac{12}{-2} \cdot -3 = -12 + 36 = 24$$

$$-26 - \frac{18}{-2} \cdot -1 = -26 + 12 = -14$$

$$0 - \frac{12}{-2} \cdot 1 = -12$$

$$-1 - \frac{-3}{-2} \cdot -3 = -1 + \frac{9}{2} = \frac{7}{2}$$

$$-4 - \frac{-1}{-2} \cdot -3 = -4 + \frac{3}{2} = -\frac{8}{2} + \frac{3}{2} = -\frac{5}{2}$$

$$9 - \frac{1}{-2} \cdot -5 = 9 - \frac{5}{2} = \frac{18}{2} - \frac{5}{2} = \frac{13}{2}$$

$$-1 - \frac{-5}{-2} \cdot -3 = -1 + \frac{15}{2} = \frac{-2}{2} + \frac{15}{2} = \frac{13}{2}$$

$$-5 - \frac{-5}{-2} \cdot -1 = -5 + \frac{5}{2} = -\frac{10}{2} + \frac{5}{2} = -\frac{5}{2}$$

$$10 - \frac{-5}{-2} \cdot 1 = 10 - \frac{5}{2} = \frac{20}{2} - \frac{5}{2} = \frac{15}{2}$$

$$0 - \frac{1}{-2} \cdot -3 = -\frac{3}{2}$$

$$0 - \frac{-1}{-2} \cdot 1 = -\frac{1}{2}$$

$$0 - \frac{1}{-2} \cdot 1 = \frac{1}{2}$$

Tableau Simplex

		x_1	x_2	x_3	x_4	x_5	x_6
z	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0	0	$\frac{1}{2}$
x_1	20	1	$\frac{1}{2}$	$\frac{1}{2}$	0	1	$\frac{1}{2}$
x_2	40	0	$\frac{1}{2}$	$\frac{1}{2}$	1	0	$\frac{1}{2}$
x_3	10	0	$\frac{1}{2}$	$\frac{1}{2}$	0	0	1
$-z$	-20	0	$\frac{1}{2}$	$\frac{1}{2}$	0	0	$\frac{1}{2}$

$z = 20$
 $-z = -20$
 $z = 20 + 20 = 40$

$$z = 20 - \frac{-8 \cdot \frac{1}{2}}{\frac{1}{2}} = 20 - 12 \cdot 8 = 20$$

$$z = 20 - 12 - \frac{-8 \cdot \frac{1}{2}}{\frac{1}{2}} = -12 - 8 = -20$$

$$z = \frac{15}{2} - \frac{-5 \cdot \frac{1}{2}}{\frac{1}{2}} = \frac{15}{2} - \frac{5}{2} = \frac{10}{2} = 5$$

$$z = 0 - 1 \cdot \frac{1}{2} = 1$$

$$-z_{\max} = -z(20; 0; 0)$$

$$z_{\min} = z(20; 0; 0) = 20 + 9 \cdot 0 + 10 \cdot 0 = 20$$

Answer: $z_{\min} = z(20; 0; 0) = 20$

$$Z = 4x_1 + x_2 + 5x_3 \rightarrow \max$$

$$\begin{cases} 3x_1 + 6x_2 + x_3 \geq 18 \\ x_1 + 5x_2 + 4x_3 \geq 20 \\ 3x_1 + 2x_2 + 5x_3 \geq 24 \\ x_i \geq 0, i = 1, 2, 3 \end{cases}$$

$$-Z = -4x_1 - x_2 - 5x_3 \rightarrow \max$$

$$\begin{cases} 3x_1 + 6x_2 + x_3 - x_4 = 18 \\ x_1 + 5x_2 + 4x_3 - x_5 = 20 \\ 3x_1 + 2x_2 + 5x_3 - x_6 = 24 \\ x_i \geq 0, i = 1, 6 \end{cases}$$

$$\bullet -1 \Rightarrow \left(\begin{array}{cccccc|c} -3 & -6 & -1 & 1 & 0 & 0 & -18 \\ -1 & -5 & -4 & 0 & 1 & 0 & -20 \\ -3 & -2 & -5 & 0 & 0 & 1 & -24 \end{array} \right)$$

Iteration 1:

B.N	R	x_1	x_2	x_3	x_4	x_5	x_6
x_4	-18	-3	-6	-1	1	0	0
x_5	-20	-1	-5	-4	0	1	0
x_6	-24	-3	-2	-5	0	0	1
$-Z$	0	-4	-1	-5	0	0	0
CO		$-\frac{4}{3}$	$-\frac{1}{3}$	-1			

$$-18 - \frac{3}{-1} \cdot -24 = -18 + 72 = 54$$

$$-20 - \frac{-5}{-1} \cdot -24 = -20 + 60 = 40$$

$$-24 : -2 = 12$$

Iteration 2:

	b.r.		x_1	x_2	x_3	x_4	x_5	x_6
DB	x_1	54	1	0	5	1	0	2
→	x_5	40	1	0	5	0	1	2
→	x_2	12	1	1	12	0	0	2
→	x_3	-12	5	0	5	0	0	1
CO	1111	2	1111	2	1111	1111	2	2

$$4 - \frac{-3 \cdot 1}{-2} = 4 - \frac{3}{2} = \frac{8-3}{2} = \frac{5}{2}$$

$$5 - \frac{1 \cdot -5}{-2} = 5 - \frac{5}{2} = \frac{10-5}{2} = \frac{5}{2}$$

$$0 - \frac{1 \cdot 1}{-2} = \frac{1}{2}$$

$$0 - \frac{-12 \cdot 1}{-2} = -12$$

$$-Z_{\max} = -Z(0; 12; 0)$$

$$Z_{\min} = Z(0; 12; 0) = 4 \cdot 0 + 12 + 5 \cdot 0 = 12$$

$$\text{Answer: } Z_{\min} = Z(0; 12; 0) = 12$$